

Eigenvalues and inverse problems

Finitely many unitary classes with prescribed shifted singular values

Difficulty: challenging **Importance:** interesting to the community**Status:** [Lean verified](#)**Topic:** Inverse determination from complete pseudospectral data**Rating rationale:** Generic finiteness follows from invariant theory, but the exceptional fibers resist that argument; the question measures how much complete pseudospectral data can identify a nonnormal matrix.

Negative resolution - George Stepaniants

Author: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA.**Solved negatively, 12 September 2026 (UTC).** The [Theorem](#) and [Sections 1-4 of the complete proof](#) establish a continuous family of complex 9×9 matrices with the same singular values of every complex scalar shift, with no two distinct family members unitarily similar. Each matrix has nilpotent Jordan type $(3,3,3)$. Thus no finite M_9 exists, refuting the universal finiteness statement below. This does not assert a classification of the remaining dimensions.[Proof PDF](#) · [Standalone proof TeX](#) · [Independent mathematical review](#) · [Submission and eligibility record](#).

The full proof passed a separate independent Codex-agent audit. Substantial AI assistance is disclosed; that earlier audit was informal automated review; the later Lean verification is recorded below. The proof uses an exact block determinant identity and a polynomial dimension argument, with the standard constant-rank theorem. Its finite diagnostic checks are not the proof. Fortier Bourque and Ransford retain credit for the question and generic finiteness theorem, which is compatible with this exceptional fiber.

Retained historical material. The original ratings, context, exact target, references and dated pre-resolution status-search text are preserved below. The earlier generic result is unchanged; the older references to an unresolved question describe the pre-resolution record.

Lean proof and verification evidence - 2026-09-17

The complete original target is Lean verified. `NLA.SP15.canonical_finiteness_false` in the [immutable proof](#) negates the finiteness assertion: for every M , there are $M+1$ complex 9-by-9 matrices with equal ordered singular values for every complex shift and no unitarily similar pair. The manuscript's smooth-curve and Jordan-type enhancements are outside this formal claim.

[Linux run 35229660506](#) passed all 29 exact Comparator contracts, default-kernel replay and rejection/isolation controls, with only `propext`, `Classical.choice` and `Quot.sound`. [Evidence and independent audits](#) distinguish local macOS Lean compilation from the final GitHub execution. Two full nonauthor mathematical reviews are retained.

Kernel-mode LeanCert certifies the consumed radius-1/16 box; an exact LU certificate proves determinant -1088 without subdivision. Pins: Lean 4.33.1, [Mathlib 0df444a3](#), and [LeanCert 621a43d7](#). [Project instructions](#) give the `lake build` reproduction command and source correspondence.

Formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology. Prior attribution is retained. Substantial AI assistance and scoped agent reviews are disclosed; external human review is not claimed. Later publication and upstream checks remain pending and will be recorded separately.

Context and notation

For a complex $N \times N$ matrix M , write $s_1(M) \geq \dots \geq s_N(M)$ for its singular values. Say $A \sim_{\text{sip}} B$ if

$$s_j(A - zI) = s_j(B - zI) \quad (z \in \mathbb{C}, 1 \leq j \leq N).$$

This is equality of *super-identical pseudospectral* data: it includes every singular value of every scalar shift.

Problem statement

Is it true that for every $N \geq 1$ there is an integer $M_N \geq 1$ such that every family $A_1, \dots, A_{M_N+1} \in \mathbb{C}^{N \times N}$ with $A_i \sim_{\text{sip}} A_j$ for all i, j contains a pair satisfying

$$A_j = U^* A_i U \quad \text{for some } i < j \text{ and } U^* U = I?$$

Equivalently, is every such data fiber a union of at most M_N unitary similarity classes, with a bound depending only on N ?

References

- Maxime Fortier Bourque and Thomas Ransford, *Super-identical pseudospectra*, *Journal of the London Mathematical Society* **79**(2), 511–528 (2009), **Theorem 1.4 and §6.2**. The theorem removes a closed measure-zero exceptional set; the question asks to remove this exception. [Author manuscript](#).
- Thomas Ransford, *Pseudospectra and matrix behaviour*, **Theorem 5.4 and the following discussion, pp. 10–11**. This restates the finiteness theorem and explicitly leaves the empty-exceptional-set case open. It explains why replacing algebraicity with integrality fails to prove it.
- G. Armentia, J. M. Gracia and F. E. Velasco, *Identical pseudospectra of any geometric multiplicity*, *Linear Algebra and its Applications* **436**(6), 1683–1688 (2012). Their similarity theorem for super-identical pseudospectra gives ordinary similarity, which allows nonunitary changes of basis and does not answer the displayed question.
- T. Ransford and N. Walsh, *A four-mean theorem and its application to pseudospectra*, [arXiv:2109.14472v2](#), 9 July 2022, Theorems 1.3–1.4. Polynomial-norm comparisons and similarity-conditioning bounds do not establish finiteness of the unitary classes; the related norm question is retained in [MF-24](#).

The original low-dimensional results give $M_2 = 1$ and $M_3 = 2$; $M_1 = 1$ is immediate. The unresolved question concerns exceptional data fibers in higher dimensions, rather than generic matrices.

Scope and status check

The quantifiers above are an explicit restatement of the source question about the exceptional set, with M_N counting classes instead of the source’s pigeonhole family size. The source is older: current openness rests on bounded later-literature searches, not a recent explicit reaffirmation. Searches through 2026-09-11 for super-identical pseudospectra, finite unitary classes, the exceptional set and related similarity results found no resolution. The 2012 ordinary-similarity theorem and the cited 2022 norm estimates were checked for scope. They do not remove the exceptional set.